OVER A BEZOUT DOMAIN EVERY FINITE SUBMODULE OF A FREE MODULE IS FREE AND EVERY FINITELY PRESENTED MODULE SPLITS AS FREE PLUS TORSION

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. The author writes '(details omitted)' once, at proof line 37861, for the reduction to a finite free ambient module; that phrase is kept verbatim. The two references inside the counted proof span are recorded as non-load-bearing navigation pointers.

Allowed standard prerequisites (editorial).

- Bezout domains: every finitely generated ideal is principal. the hypothesis at statement line 37848 and used at proof lines 37862-37863. Is the input for the n = 1 case and for the rank-one pieces of the induction.
- The Warfield decomposition for finitely presented modules over a ring whose localizations are valuation rings. invoked at proof lines 37873-37876. Produces the ambient module R^{oplus r} oplus bigoplus R/f_iR of which M is a summand.
- Functoriality of the torsion submodule and splitting of a direct summand sequence. used at proof lines 37877-37883. Separates the free and torsion parts and assembles the direct sum decomposition of M.

Source reference. Original lemma 37834: sources/stacks-more-algebra/pinned-original-source.tex, source lines 37834–37834. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 37737: sources/stacks-more-algebra/pinned-original-source.tex, source lines 37737–37737. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Lemma 0.2. *Let R be a Bézout domain.*

(1) *Every finite submodule of a free module is finite free.*

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(2) *Every finitely presented R-module M is a direct sum of a finite free module and a torsion module M_{tors} which is a summand of a module of the form $\bigoplus_{i=1,\ldots,n} R/f_iR$ with $f_1, \ldots, f_n \in R$ nonzero.*

Proof. Proof of (1). Let $M \subset F$ be a finite submodule of a free module F. Since M is finite, we may assume F is a finite free module (details omitted). Say $F = R^{\oplus n}$. We argue by induction on n. If $n = 1$, then M is a finitely generated ideal, hence principal by our assumption that R is Bézout. If $n > 1$, then we consider the image I of M under the projection $R^{\oplus n} \to R$ onto the last summand. If $I = (0)$, then $M \subset R^{\oplus n-1}$ and we are done by induction. If $I \neq 0$, then $I = (f) \cong R$. Hence $M \cong R \oplus \mathrm{Ker}(M \to I)$ and we are done by induction as well.

Let M be a finitely presented R-module. Since the localizations of R are maximal ideals are valuation rings (Lemma 37834) we may apply Lemma 37737. Thus M is a summand of a module of the form $R^{\oplus r} \oplus \bigoplus_{i=1,\ldots,n} R/f_iR$ with $f_i \neq 0$. Since taking the torsion submodule is a functor we see that M_{tors} is a summand of the module $\bigoplus_{i=1,\ldots,n} R/f_iR$ and M/M_{tors} is a summand of $R^{\oplus r}$. By the first part of the proof we see that M/M_{tors} is finite free. Hence $M \cong M_{tors} \oplus M/M_{tors}$ as desired. □